ARRAYLIST

CCE 103/L

OBJECTIVES

1. Understands the purpose and benefits of using ArrayLists compared to traditional arrays.
2. Declare, create, and initialize ArrayList of different data types.
3. Access and modify elements based on their index.
4. Use method add(), set(), remove(), clear() and size()

ARRAYLIST

- Class is resizable array, which can be found in the java.util package.
- It follows the same rules as Arrays when it comes to indexing
- It doesn’t use square brackets.

INITIALIZING ARRAYLIST

- **Import Arraylist**
  - Import java.util. ArrayList
- **Declare ArrayList**
  - Arraylist <datatype> identifier = new ArrayList<datatype>();

- Note: Arraylist uses a Non-Primitive Data Type

INITIALIZING ARRAYLIST

```java
package arraylists;
import java.util.ArrayList;

public class ArrayLists {

    /**
     * @param args the command line arguments
     */
    public static void main(String[] args) {

        ArrayList <String> name = new ArrayList<String>();

    }

}
```

1- import arraylist

2 – declaring arraylist

ADD ITEM IN ARRAYLIST

- To add elements in arraylist use add() method.
- ✓ **identifier.add(“”);**

```java
package arraylists;
import java.util.ArrayList;


public class ArrayLists {

    /**
     * @param args the command line arguments
     */
    public static void main(String[] args) {

        ArrayList <String> name = new ArrayList<String>();
        name.add("David");
        name.add("Carlo");


    }
```

ACCESS AN ITEM

- To access an element in the Arraylist, use the get() method and refer to the index number.
- ✓ **identifier.get(index);**

```
public class ArrayLists {

    public static void main(String[] args) {

        ArrayList <String> name = new ArrayList<String>();
        name.add("David");
        name.add("Carlo");
        name.add("Julios");
        name.add("Nathan");

        String student = name.get(1);
        System.out.println(student);
        System.out.println(name.get(0));
    }
}
```

CHANGE ITEM IN ARRAYLIST

- To update an element in arraylist, use set() method and refer to the index number.

✓ **identifier.set(index, value);**

```
public class ArrayLists {

    public static void main(String[] args) {

        ArrayList <String> name = new ArrayList<String>();
        name.add("David");
        name.add("Carlo");
        name.add("Julios");
        name.add("Nathan");

        name.set(0,"John");
        System.out.println(name.get(0));
    }
}
```

REMOVE ITEM IN ARRAYLIST

- To remove an element in arraylist, use remove() method and refer to the index number.
  - ✓ **identifier.remove(index);**
- To remove all element in arraylist, use clear() method.
  - ✓ **identifier.clear(index);**

REMOVE ITEM IN ARRAYLIST

```java
public class ArrayLists {

    public static void main(String[] args) {

        ArrayList <String> name = new ArrayList<String>();
        name.add("David");
        name.add("Carlo");
        name.add("Julios");
        name.add("Nathan");

        name.remove(0);
        System.out.println(name.get(0));
    }
}
```

REMOVE ITEM IN ARRAYLIST

```java
public class ArrayLists {

    public static void main(String[] args) {

        ArrayList <String> name = new ArrayList<String>();
        name.add("David");
        name.add("Carlo");
        name.add("Julios");
        name.add("Nathan");

        name.clear();
        System.out.println(name.get(0));
    }
}
```

Output

total (run) × ArrayLists (run) ×

```
run:
Exception in thread "main" java.lang.IndexOutOfBoundsException: Index: 0, Size: 0
        at java.util.ArrayList.rangeCheck(ArrayList.java:653)
        at java.util.ArrayList.get(ArrayList.java:429)
        at arraylists.ArrayLists.main(ArrayLists.java:21)
C:\Users\Kates\AppData\Local\NetBeans\Cache\8.1\executor-snippets\run.xml:53: Java returned: 1
BUILD FAILED (total time: 0 seconds)
```

ARRAYLIST SIZE

- To find out how many elements an Arraylist have, use size() method.
  - **identifier.size(**

```
public class ArrayLists {

    public static void main(String[] args) {

        ArrayList <String> name = new ArrayList<String>();
        name.add("David");
        name.add("Carlo");
        name.add("Julios");
        name.add("Nathan");

        int arr_size= name.size();
         System.out.println(arr_size);
        System.out.println(name.size());
    }
}
```

ITERATING ARRAYLIST

Loop through the elements of an arraylist with for loop, and use the size() method to specify how many times the loop should run:

```java
public class ArrayLists {

    public static void main(String[] args) {

        ArrayList <String> name = new ArrayList<String>();
        name.add("David");
        name.add("Carlo");
        name.add("Julios");
        name.add("Nathan");

        for(int i=0; i<name.size();i++){
            System.out.println(name.get(i));
        }
    }
}
```

Using for loop

ITERATING ARRAYLIST

```java
public class ArrayLists {

    public static void main(String[] args) {

        ArrayList <String> name = new ArrayList<String>();
        name.add("David");
        name.add("Carlo");
        name.add("Julios");
        name.add("Nathan");

      for(String names:name){
          System.out.println(names);
      }
    }
}
```

Using for each loop